

Principles of Economics

Problem Set 7

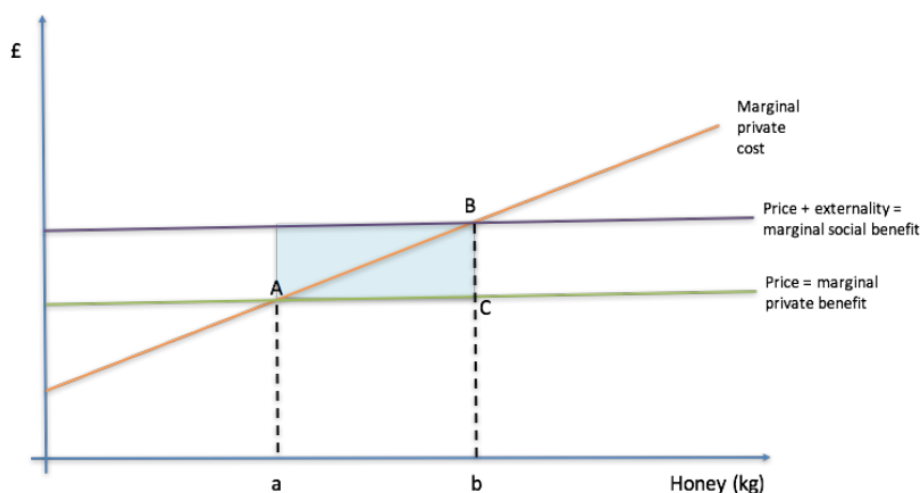
Universidad Carlos III de Madrid | Department of Economics

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1. WARMING-UP QUESTIONS

1. Imagine a beekeeper, who produces honey and sells it at a constant price per kilogram.
 - a. Draw a diagram with the quantity of honey on the horizontal axis, showing the marginal cost of honey production as an upward-sloping line, and the price of honey as a horizontal line. Show the amount of honey that the profit-maximizing beekeeper will produce.
 - b. For the beekeeper, the marginal private benefit of producing a kilo of honey is equal to the price. But since the bees benefit a neighbouring farmer, by helping to pollinate her crops, honey production has a positive external effect. Draw a line on your diagram to represent the marginal social benefit of honey production. Show the quantity of honey that would be Pareto efficient. How does it compare with the quantity chosen by the beekeeper?
 - c. Explain how the farmer and beekeeper could both be made better off through bargaining. What is the equilibrium price, quantity, producer and consumer surplus in equilibrium?

Exercise 12.3 A Positive Externality



2. Construct a table like the one in Figure 12.13 to analyse the possible market failures associated with the decisions below. In each case, can you identify which markets or contracts are missing or incomplete?
 - a. You inoculate your child with a costly vaccination against an infectious disease.

- b. You use money that you borrow from the bank to invest in a highly risky project.
- c. A fishing fleet moves from the overfished coastal waters of its own country to international waters.
- d. A city airport increases its number of passenger flights by allowing nighttime departures.
- e. You contribute to a Wikipedia page.
- f. A government invests in research in nuclear fusion.

The decision	How it affects others	Cost or benefit	Market failure	Possible remedies	Terms applied to this type of market failure
You inoculate your child with a costly vaccination against an infectious disease	Others likely to catch the disease	Private cost, external benefit	Amount of vaccination is below social optimum	Mandatory vaccination; Subsidized or government-provided vaccinations	Positive external effect
You use money that you borrow from the bank to invest in a highly risky project	The bank may make a loss if the project is unsuccessful.	Private benefit, external cost	Risky projects are more likely to be financed by bank loans. Low effort levels	Loans based on previous credit checks, project-based lending with personal liability.	Incomplete contracts, imperfect information, moral hazard,
A fishing fleet moves from the overfished coastal waters of its own country to international waters	Fishermen in international waters have fewer fish to catch	Private benefit, external cost	Overfishing	Quotas on the fish caught	Negative external effect, environmental spillovers, common pool resource.
A city airport increases its number of passenger flights by allowing night-time departures	Adverse effects on inhabitants living near the airport due to the noise and pollution	External cost, private benefit	Overuse of air travel	Regulation of night-time departures	Negative external effect, environmental spillovers
You contribute to a Wikipedia page	Others can read the page and benefit from your knowledge. No quality control may mean that low quality articles make it online.	External benefit, private cost (in terms of time)	Free-riding on contribution of others. Low engagement of publishers.	Payments in order to access the article. Contribution requirement to be able to read.	Positive external effects. Moral hazard.
A government invests in research in nuclear fusion	Others can exploit the research	Private cost for governments (tax payers), public benefit.	Free-riding on the research effort of one country.	Payments for use of the technology.	Positive external effects

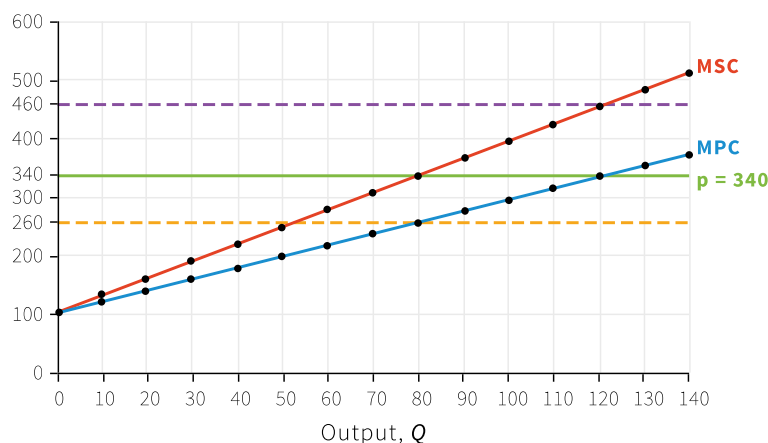
2. MULTIPLE-CHOICE QUESTIONS

3. Read the following statements and choose the correct option(s).
- a) Some public goods are rival.
 - b) A public good must be non-excludable.
 - c) A good cannot be rival and non-excludable.
 - d) If a good is non-rival, then the cost of an additional person consuming it is zero.
4. Read the following statements and choose the correct option(s).
- a) Market failure occurs in the credit market because loans are given to rich people with low-quality projects.
 - b) It is easier for rich people to get loans because they are able to provide equity or collateral.
 - c) Hidden action can lead to markets being missing altogether.
 - d) Full insurance can mitigate the principal-agent problem in insurance markets.
5. In which of the following cases is there an adverse-selection problem?
- a) A motor insurance market, in which the insurers do not know how carefully the insured people drive.
 - b) A health insurance market, in which the insurers do not know whether or not the applicants for insurance are habitual smokers.
 - c) Online sales of nutritional supplements, when consumers cannot tell whether their contents are as claimed by sellers.
 - d) A firm that employs home workers, but cannot observe how hard they are working.
6. Consider two countries: CarWise and Haven'tGotAClue. In CarWise, the second-hand car salesmen know the values of the cars that they are selling, while in Haven'tGotAClue they only know the distribution of the quality of the cars - how many cars are of which quality, but not the quality of any particular car. The buyers only know the quality distribution in both countries. Based on this information, which of the following statements is correct?
- a. In both countries, the second-hand car market suffers from adverse selection.
 - b. All cars will be sold in CarWise.
 - c. No cars will be sold in Haven'tGotAClue.
 - d. In CarWise, only the worst quality cars will be offered for sale.

3. NUMERICAL PROBLEMS

7. A factory is situated next to a dormitory for nurses who work night shifts. The factory produces 120 units of humanoid robots a day. The production process is rather noisy, and the nurses often complain that their sleep is disturbed. The following graph depicts the MPC

and MSC of the robot factory production. The robot market is competitive and the market price is £340. Based on this information, which of the following statements is/are correct?



- The factory's profit-maximising quantity is 80.
 - The efficient level of output is 0, where there is zero marginal external cost to the nurses.
 - At $Q = 120$, both the factory and the nurses would benefit from the nurses paying a fee less than 120 to the factory to reduce output.
 - At $Q = 80$, both the factory and the nurses would benefit from the factory paying a fee less than 80 to the nurses to increase output.
8. The graph in the previous question depicts the MPC and MSC of the robot factory production. The robot market is competitive and the market price is £340. The local authority has now passed a law that ensures that the nurses have the right to undisturbed sleep (meaning zero production). Based on this information, which of the following statements are correct?
- The Pareto efficient output is 120.
 - The nurses' minimum acceptable payment from the factory to produce the efficient outcome is £4,000.
 - The maximum that the factory is willing to pay to increase the output to the efficient level is £6,400.
 - The net social gain from attaining the efficient output level is £9,600.
9. Consider again the factory and the nurses. Instead of a Pigouvian tax, the government is now considering implementing an enforced compensation from the factory to the nurses for the costs imposed. Based on this information, which of the following statements is correct?
- The factory's marginal private cost now equals the marginal social cost.
 - The total compensation paid will be £6,400.
 - The nurses are worse off than they would be under the Pigouvian tax.
 - The factory's profit would be the same as under the Pigouvian tax.

4. CONCEPTUAL QUESTIONS

10. Consider the policy regulation to solve externality problems. Evaluate the strengths and weaknesses of each policy from the standpoint of Pareto efficiency and fairness.
11. Consider the policy of Pigouvian tax to solve externality problems. Evaluate the strengths and weaknesses of each policy from the standpoint of Pareto efficiency and fairness.
12. In two cases described in the book about externality (incinerator, loud music) explain why the external effects are not (and possibly cannot be) covered by a complete contract. Explain also what critical piece(s) of information required for a complete contract are asymmetric or non-verifiable?